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THE KNOWLEDGE UNIVERSITY

06 – Routing

IE2004 Computer Networks

2026 July

Lesson outline

- Routing and Forwarding
- Routing table
- Longest prefix match
- Static routing vs Dynamic Routing
- Dynamic routing protocol classification
- Behavior of Distance Vector and Link State protocols

Routing vs Forwarding

network-layer functions:

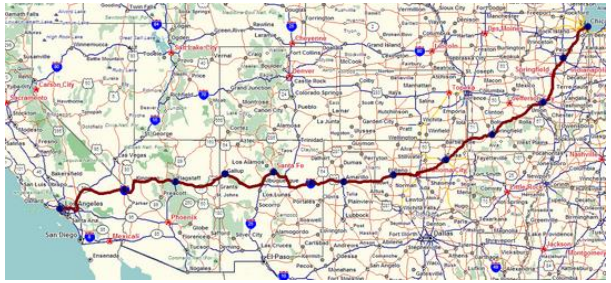
- *forwarding:* move packets from a router's input link to appropriate router output link
- *routing:* determine route taken by packets from source to destination
  - *routing algorithms*

analogy: taking a trip

- *forwarding:* process of getting through a single interchange or junction
- *routing:* process of planning a trip from source to destination



forwarding



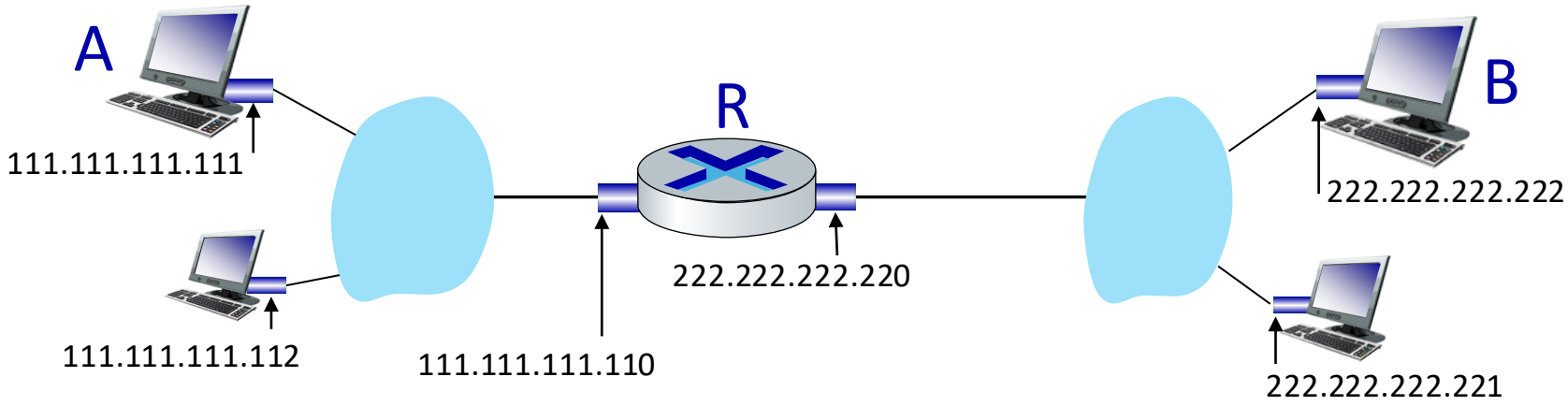
routing

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Routing

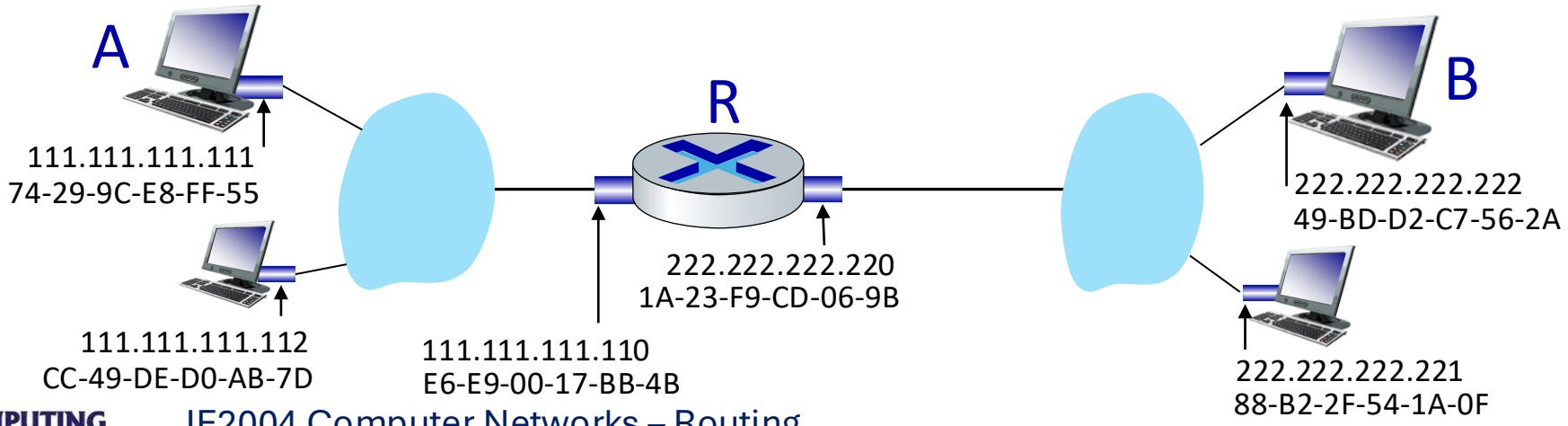
- A router interconnects two or more networks at Layer 3
- In this example, we have two networks 111.111.111.0/24 and 222.222.222.0/24
- The hosts A and B cannot talk to each other directly
- IP datagrams between A and B must go through the router R
- How does the router R forward an IP datagram from A to B?



Routing to another subnet: addressing

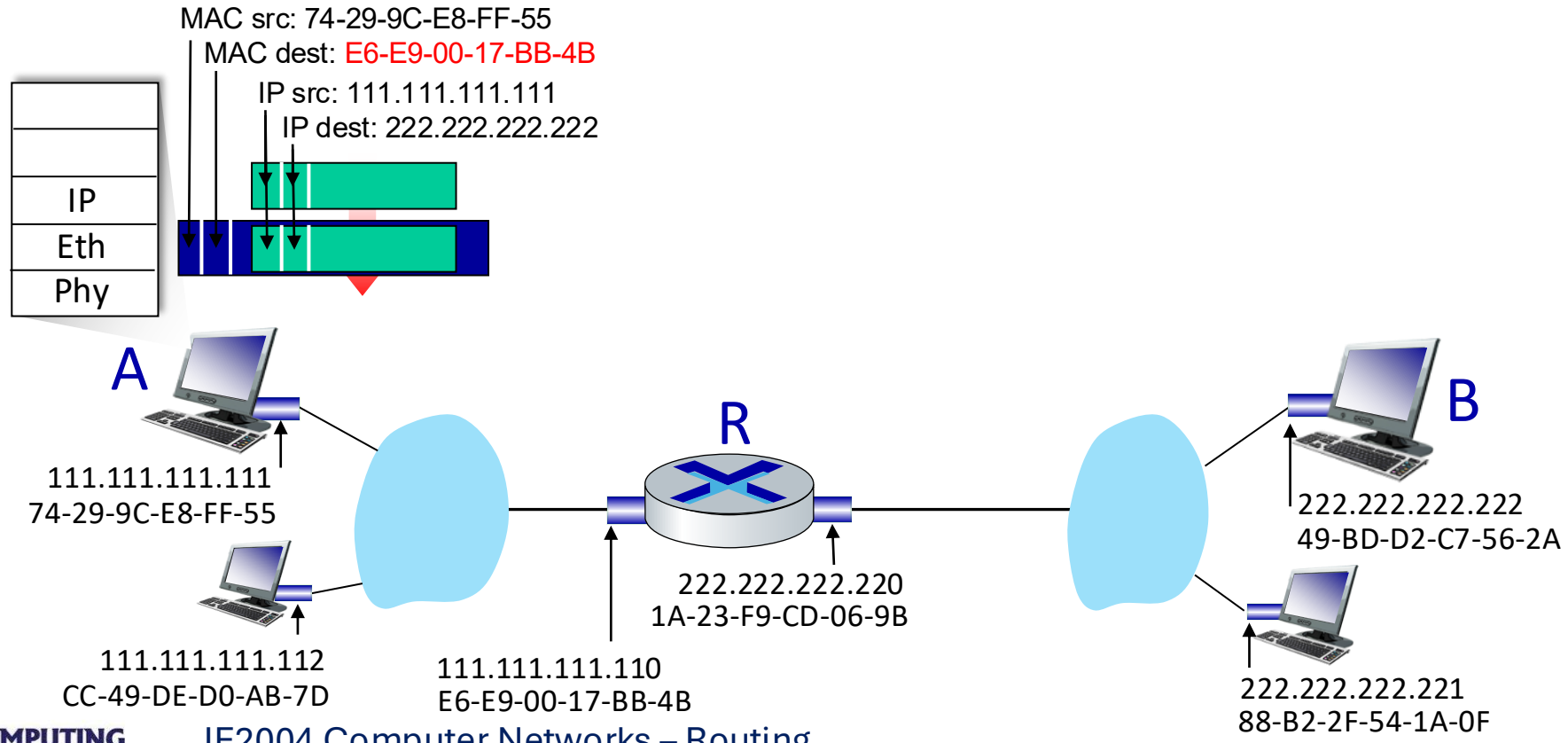
walkthrough: sending a datagram from *A* to *B* via *R*

- focus on addressing – at IP (datagram) and MAC layer (frame) levels
- assume that:
  - A knows B's IP address
  - A knows IP address of first hop router, R (how?)
  - A knows R's MAC address (how?)



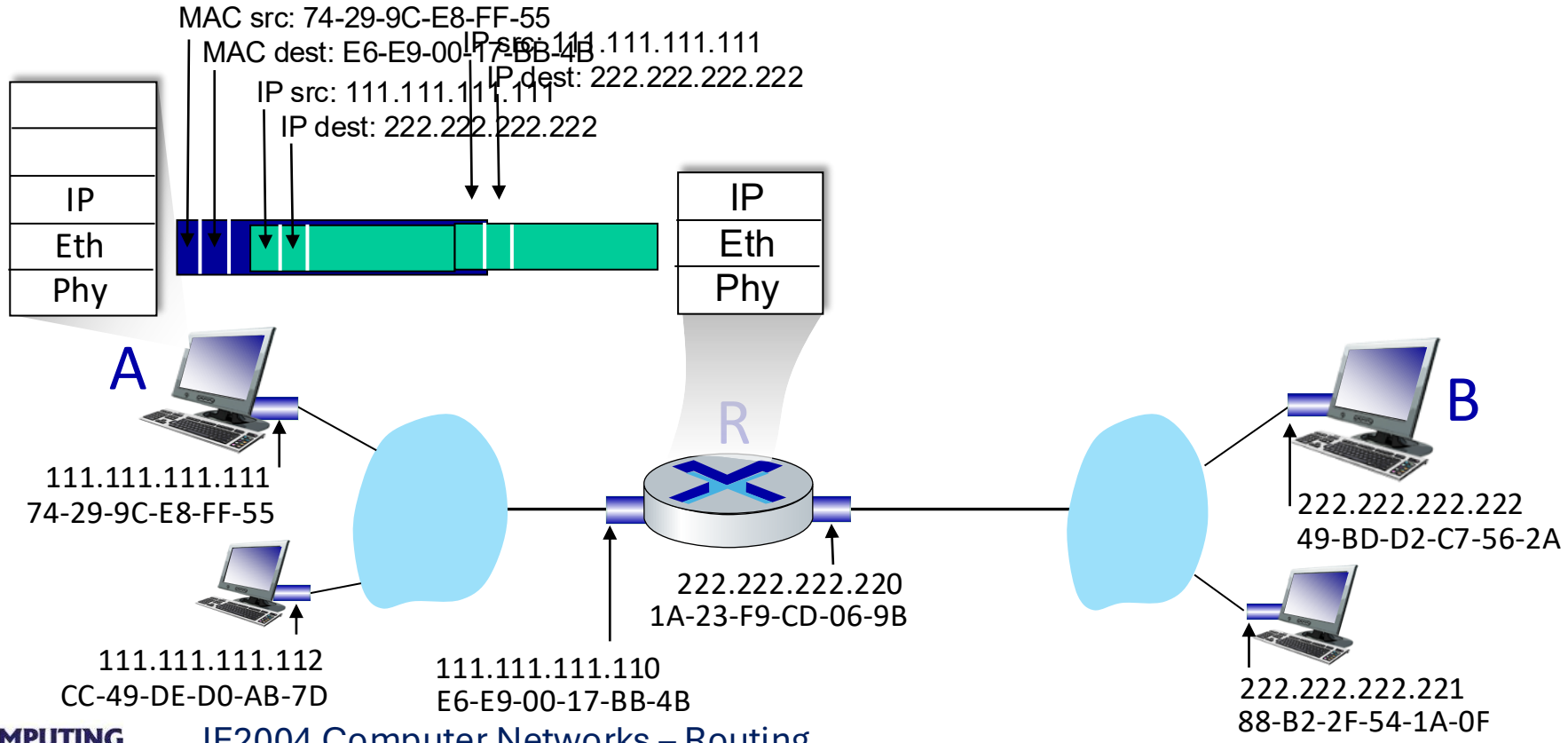
Routing to another subnet: addressing

- A creates IP datagram with IP source A, destination B
- A creates link-layer frame containing A-to-B IP datagram
  - R's MAC address is frame's destination



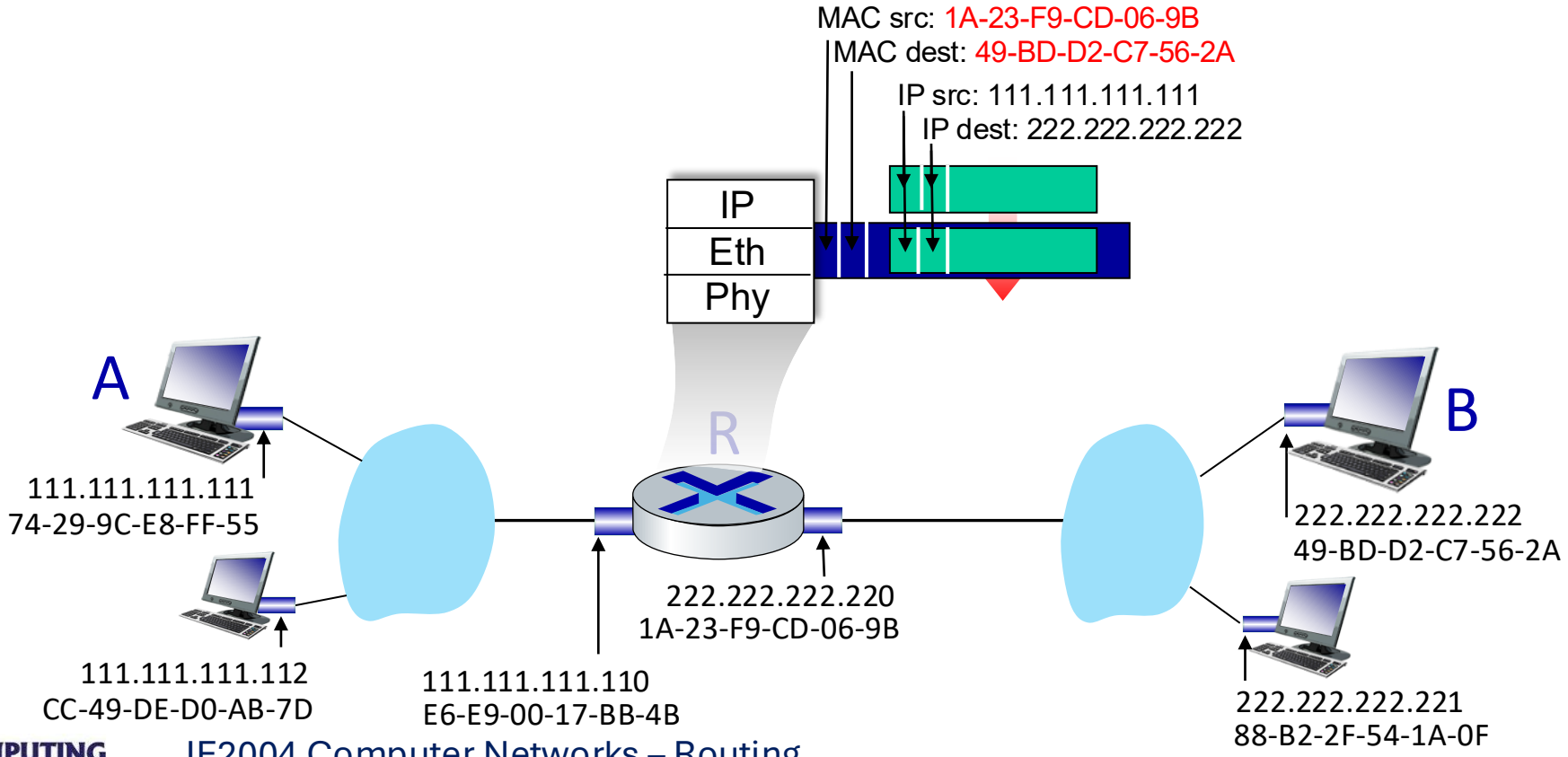
Routing to another subnet: addressing

- frame sent from A to R
- frame received at R, datagram extracted, passed up to IP



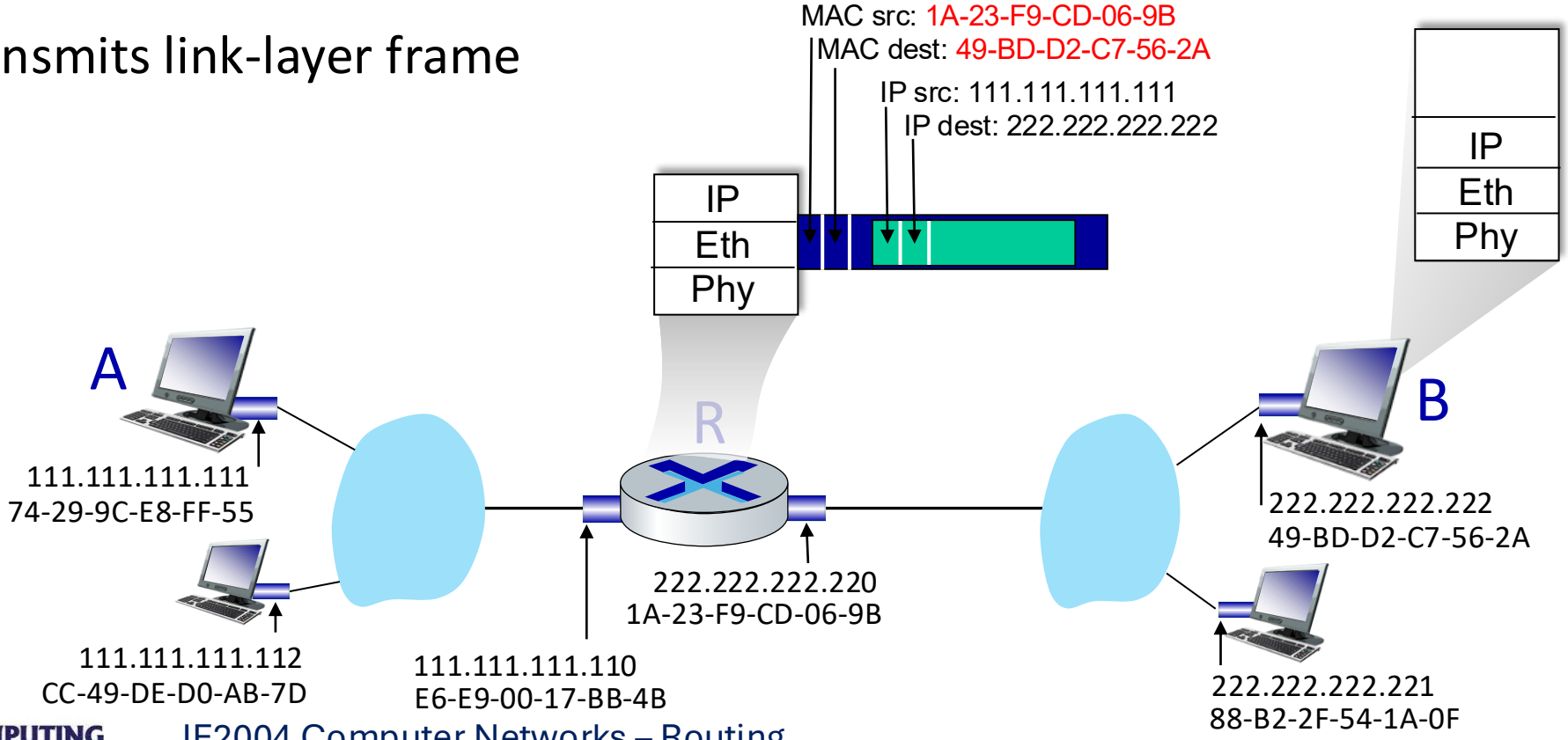
Routing to another subnet: addressing

- R determines outgoing interface, passes datagram with IP source A, destination B to link layer
- R creates link-layer frame containing A-to-B IP datagram. Frame destination address: B's MAC address



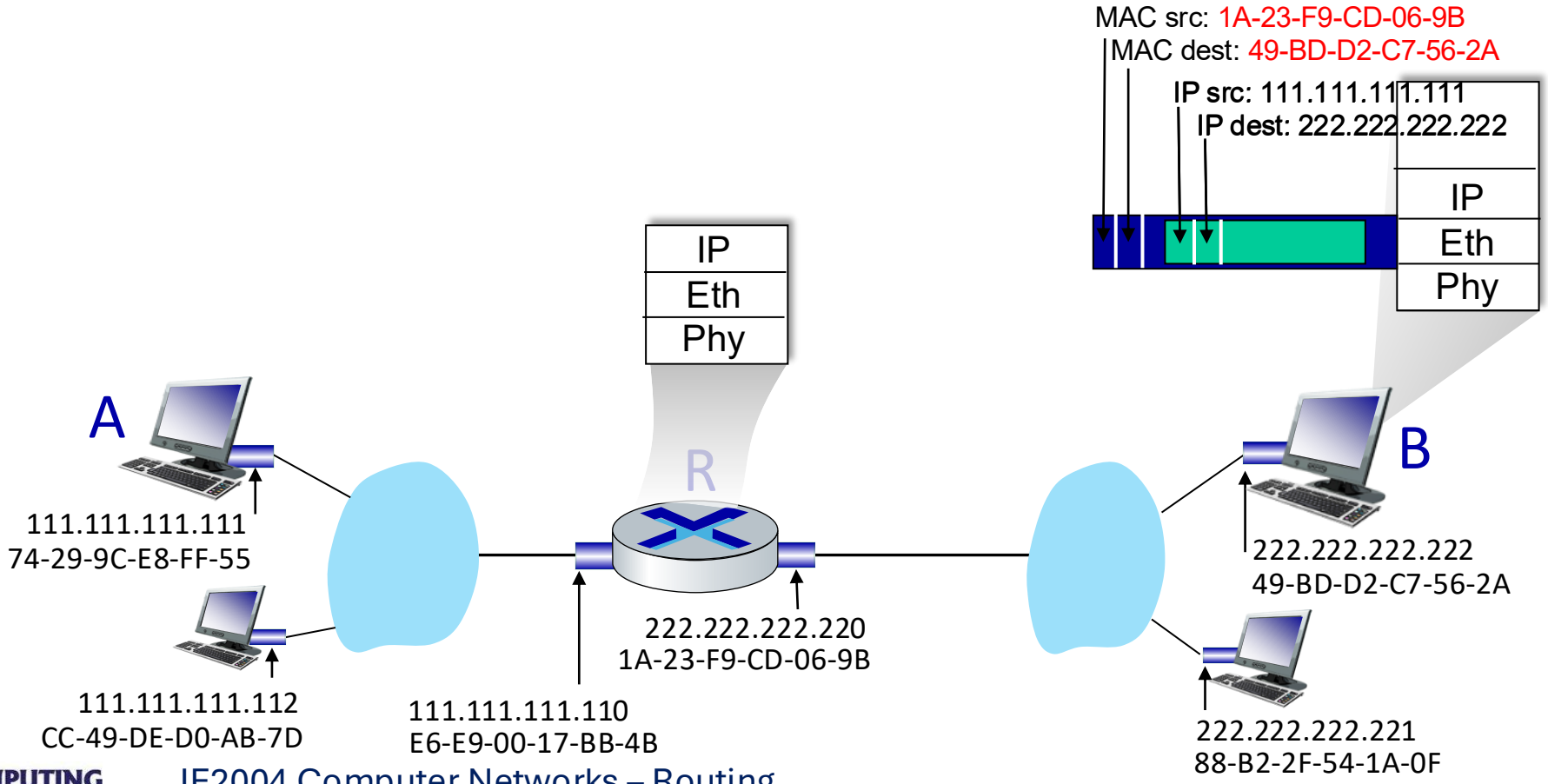
Routing to another subnet: addressing

- R determines outgoing interface, passes datagram with IP source A, destination B to link layer
- R creates link-layer frame containing A-to-B IP datagram. Frame destination address: B's MAC address
- transmits link-layer frame



Routing to another subnet: addressing

- B receives frame, extracts IP datagram destination B
- B passes datagram up protocol stack to IP

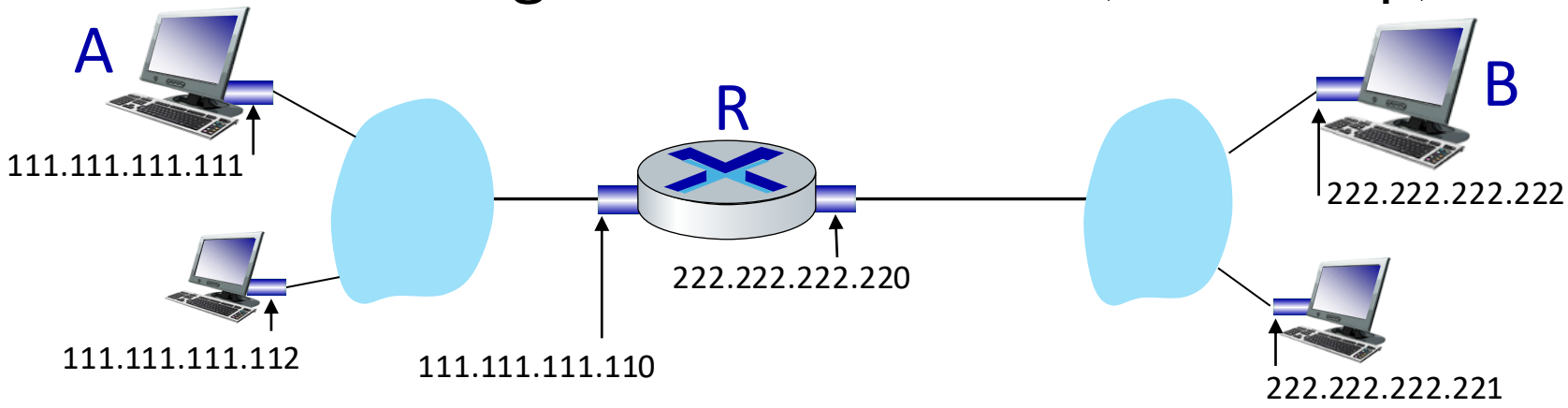


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Routing table

- In this example, the router R's task was easy, because destination IP address 222.222.222.222 was on a directly connected network
- In general, routers have to do a bit more work to determine how to forward the datagram
- A router maintains a Routing Table that it can look up
  - Actually, even a host must have a routing table (usually with a few rows)
- Columns in the Routing Table: Destination, Next Hop, Interface, …



Routing tables

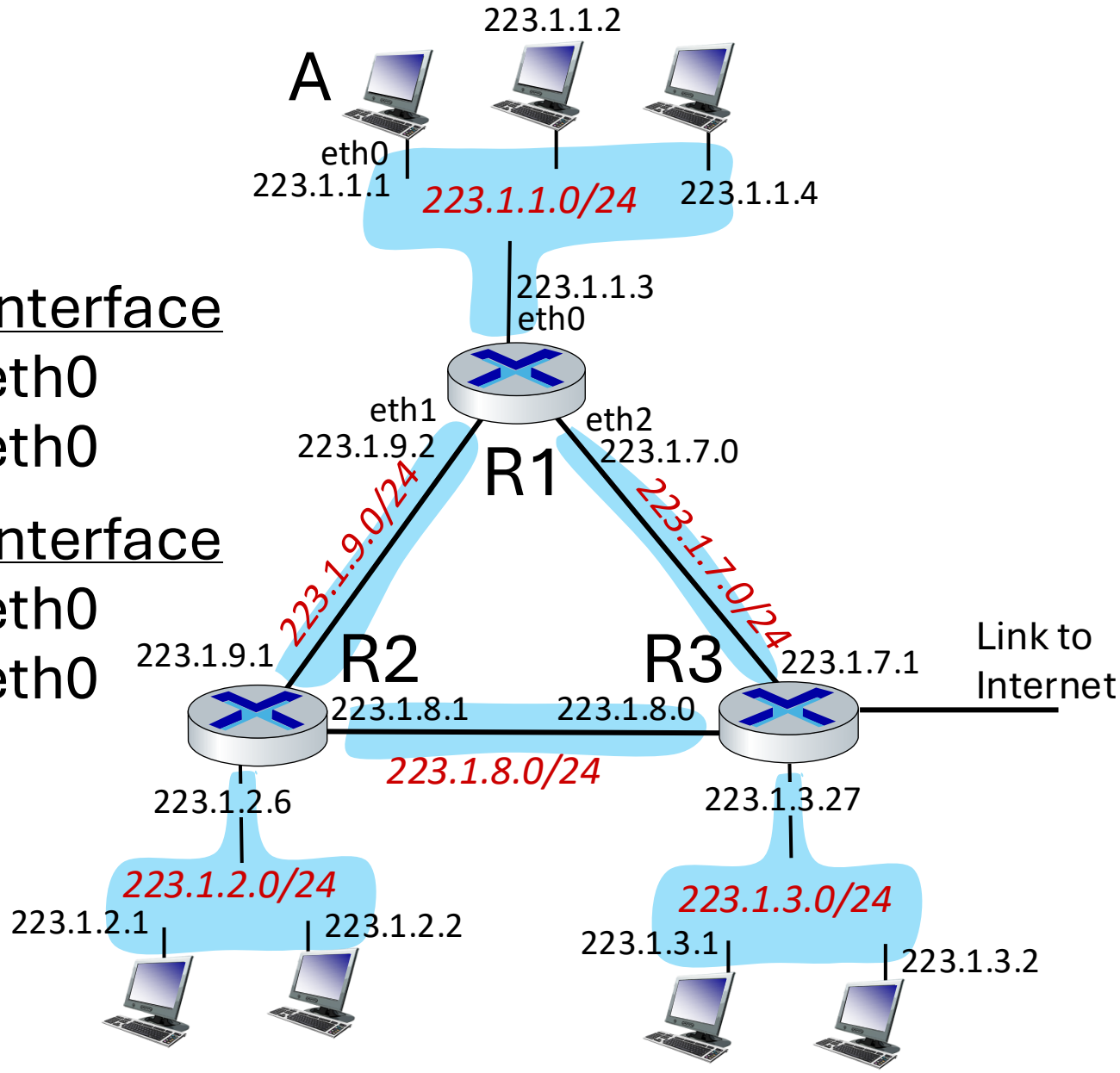
- Routing table at Host A

Destination	Next Hop	Interface
223.1.1.0/24	Connected	eth0
default	223.1.1.3	eth0

- Destination / Next Hop / Interface

Destination	Next Hop	Interface
223.1.1.0/24	Connected	eth0
0.0.0.0/0	223.1.1.3	eth0

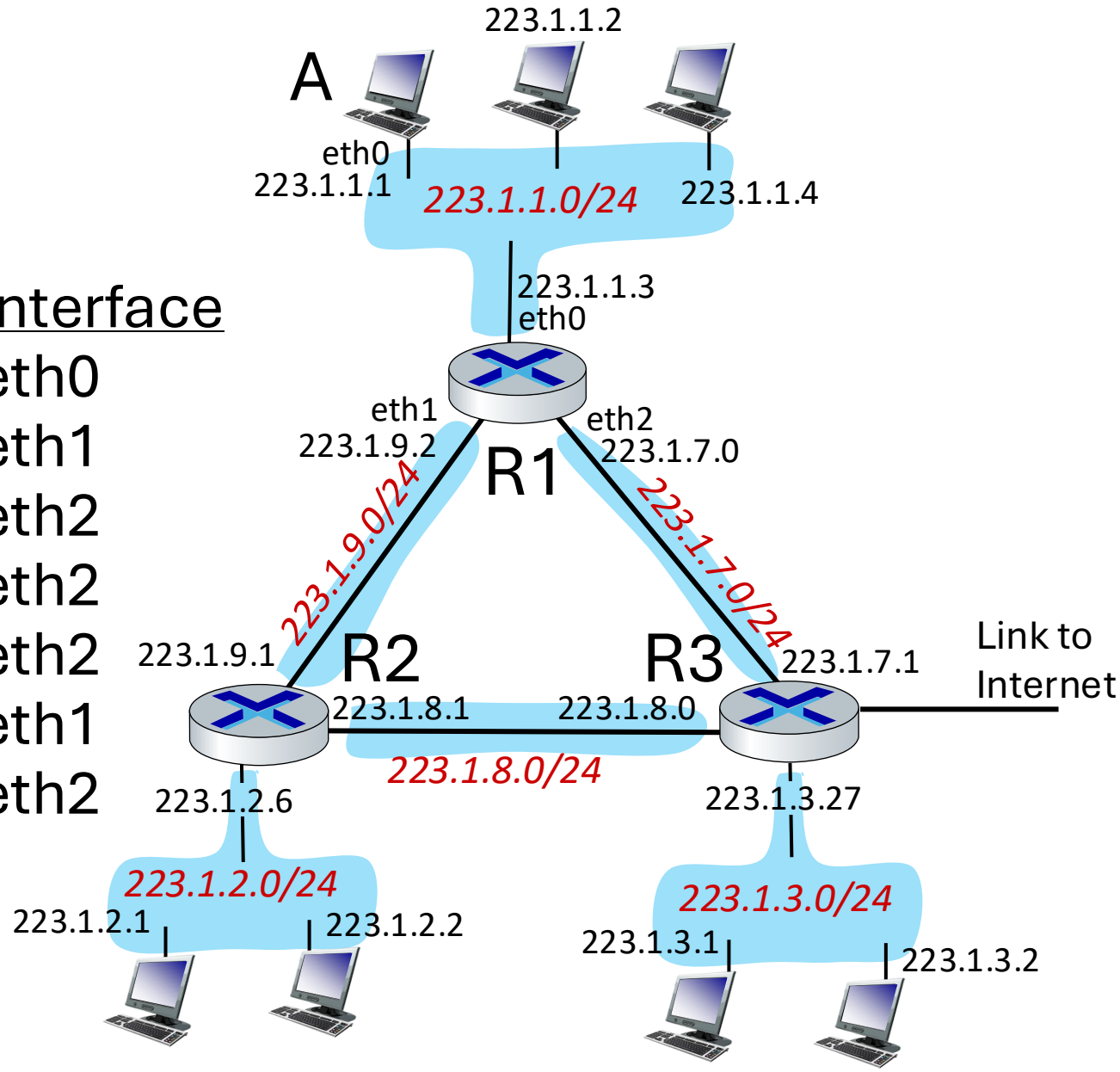
- default = 0.0.0.0/0



Routing tables

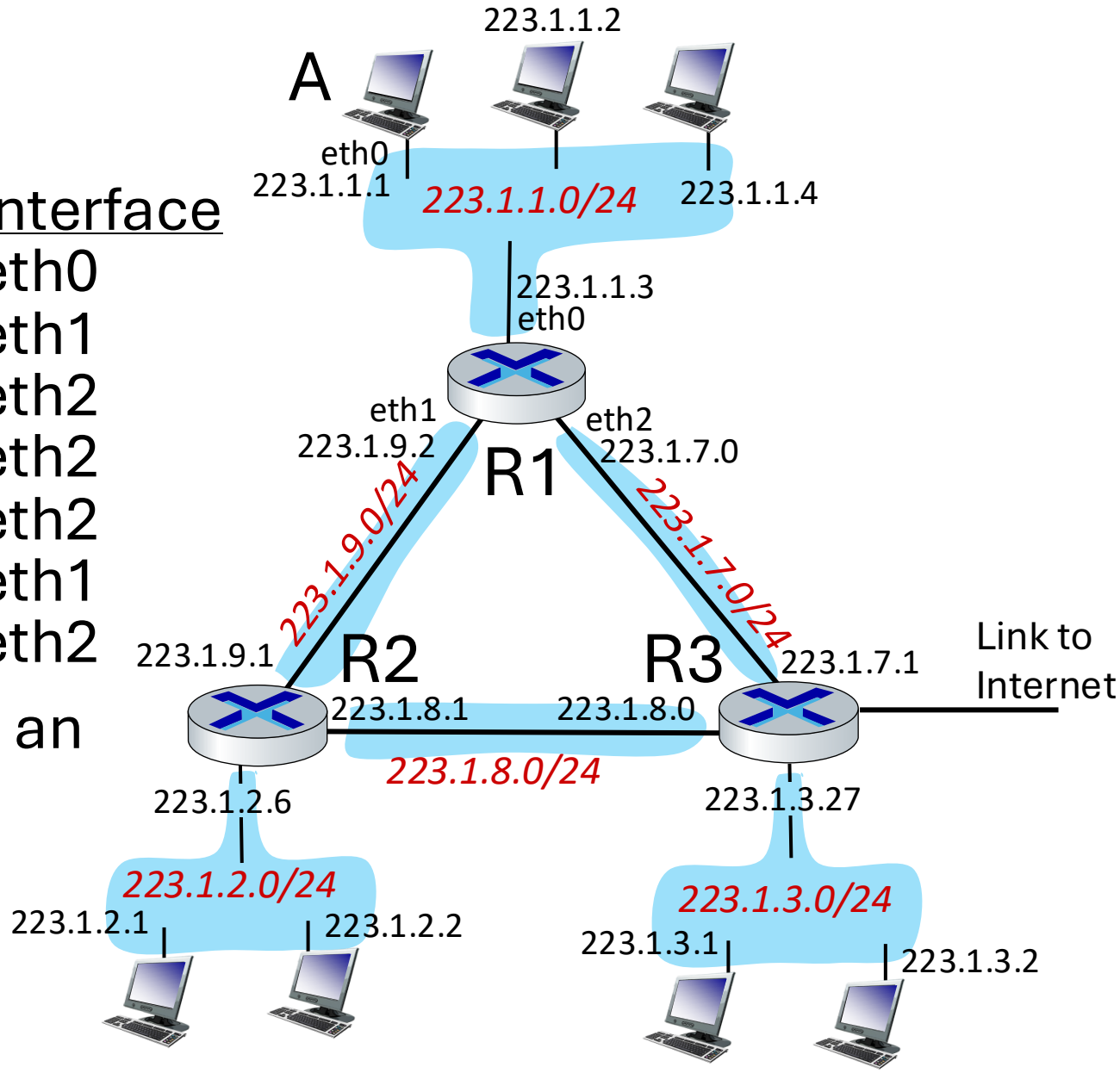
- Routing table at Router R1
-

Destination	Next Hop	Interface
223.1.1.0/24	Connected	eth0
223.1.2.0/24	223.1.9.1	eth1
223.1.3.0/24	223.1.7.1	eth2
223.1.7.0/24	Connected	eth2
223.1.8.0/24	223.1.7.1	eth2
223.1.9.0/24	Connected	eth1
default	223.1.7.1	eth2



Routing at R1

Destination	Next Hop	Interface
223.1.1.0/24	Connected	eth0
223.1.2.0/24	223.1.9.1	eth1
223.1.3.0/24	223.1.7.1	eth2
223.1.7.0/24	Connected	eth2
223.1.8.0/24	223.1.7.1	eth2
223.1.9.0/24	Connected	eth1
default	223.1.7.1	eth2
- What should R1 do if it receives an IP datagram from A destined to
  - 223.1.2.1 ?
  - 223.1.3.2 ?
  - 8.8.8.8 ?



Exercises

- Determine the routing tables at R2 and R3 (like slide 6-16)

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Routing process at a router

- Compare the destination IP address against ALL entries in the routing table
- Select the Next Hop / Interface that results in the longest prefix match with the destination IP address
- For an address to match with an entry W.X.Y.Z/P the first P bits of the address must match with the first P bits of W.X.Y.Z
- Examples:
  - 223.1.3.2 does not match with 223.1.2.0/24 because only the first 23 bits match
  - 223.1.2.2 matches with 223.1.2.0/24 because all of the first 24 bits match

Longest prefix match - examples

- Consider the following routing table at a router

	Destination				Next hop	Interface
200.23.16.0/21	11001000	00010111	00010***	********	R1	eth0
200.23.24.0/21	11001000	00010111	00011***	********	R2	eth1
200.23.24.0/24	11001000	00010111	00011000	********	R3	eth2
0.0.0.0/0	********	********	********	********	R4	eth3

- Which entry would have the longest prefix match for the following destination IP addresses?

200.23.16.5	11001000	00010111	00010000	00000101	R1	eth0
200.23.26.1	11001000	00010111	00011010	00000001	R2	eth1
200.23.24.4	11001000	00010111	00011000	00000100	R3	eth2
8.8.8.8	00001000	00001000	00001000	00001000	R4	eth3

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Static vs Dynamic Routing

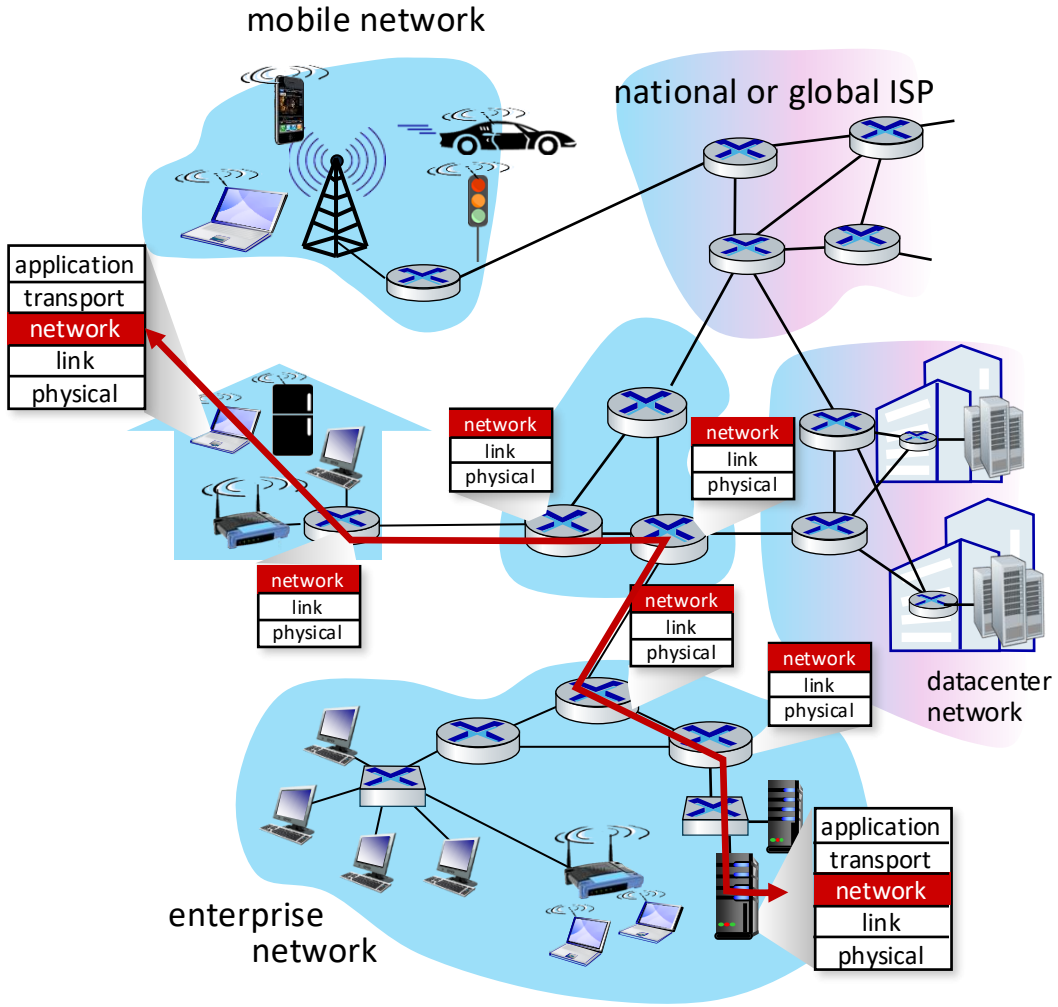
- How do the entries in Routing tables get created?
  - Static Routing: An administrator enters them manually
  - Dynamic Routing: Routers exchange routing information and learn routes dynamically
- Static routing
  - Fixed path from a given network to another
  - Suitable only for small networks (i.e. 3 or 4 routers)
  - Not suitable for large networks
    - Time consuming to manually configure static routes on all the routers
    - Routers and links can go down or links can become congested, and packets may need to be routed via alternate paths depending on the status of the network

Dynamic Routing

- Routers exchange routing information using a Routing Protocol
- Routing protocol characteristics
  - What routing information is exchanged?
  - With which routers does a particular router exchange routing information?
  - How often are the routing information exchanged?
- Protocol classification
  - Distance vector protocols vs Link state protocols
  - Interior gateway protocols vs Exterior gateway protocols

Dynamic Routing protocols

- Routing protocol goal: determine "good" paths (routes), from a sending hosts to receiving host, through network of routers
- path: sequence of routers packets traverse from given initial source network to final destination network
- "good": least "cost", "fastest", "least congested"
- Routing is a major networking challenge!



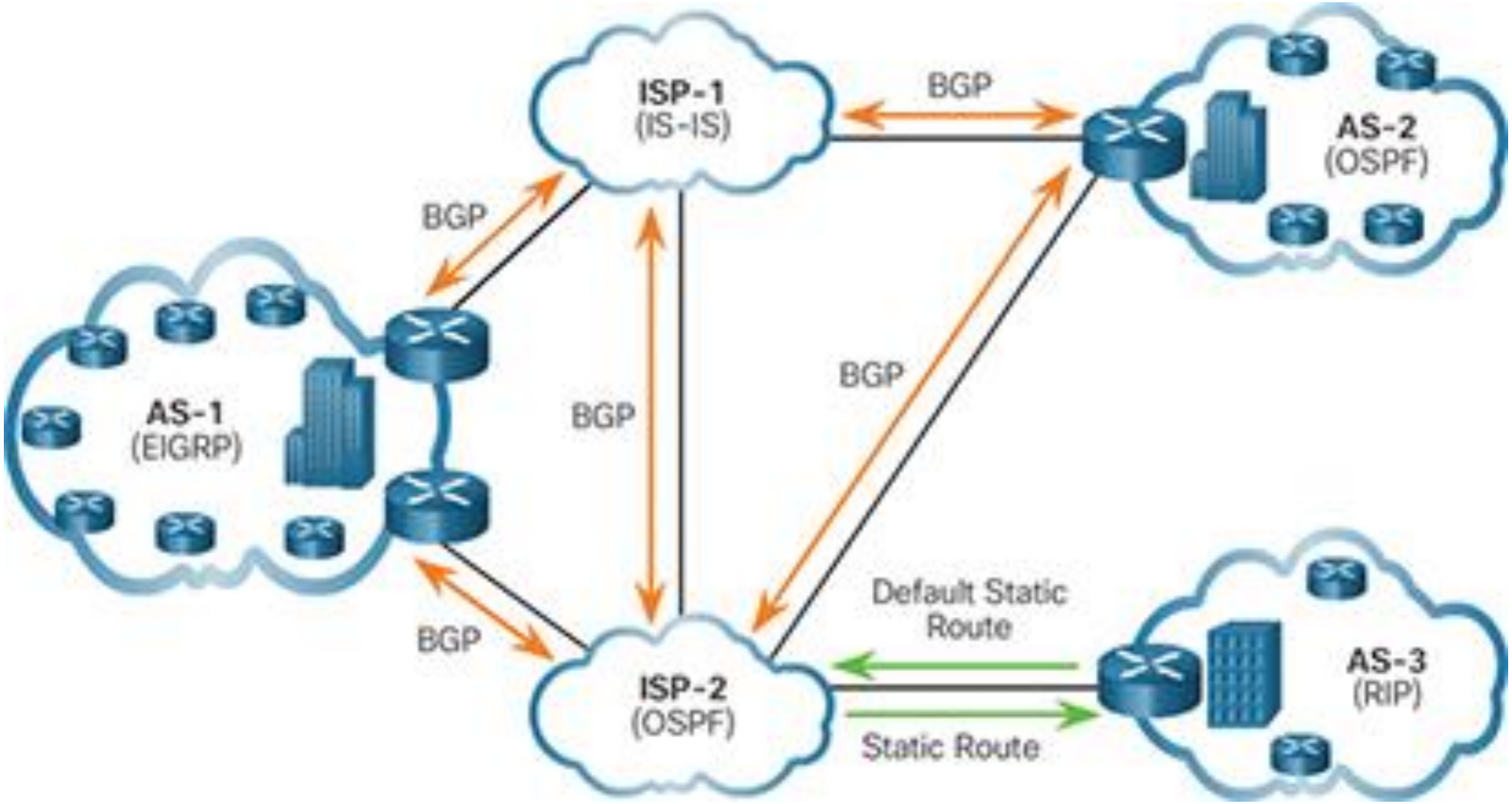
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Autonomous systems

- Typically, an organization would have several routers within its network and at least one connection to the outside world
- A group of networks and routers controlled by a single administrative entity is called an Autonomous System (AS)
- An autonomous system is connected to the rest of the world via one or many *gateways*
  - A gateway is a router within an AS with a link to the outside world
- Interior Gateway Protocols
  - Routing protocols used for exchange of routing information within an AS
  - Examples: RIP, OSPF, IS-IS, EIGRP (Cisco)
- Exterior Gateway Protocols
  - Routing protocols used for exchange of routing information between ASs
  - Examples: BGP, EGP

Autonomous systems example

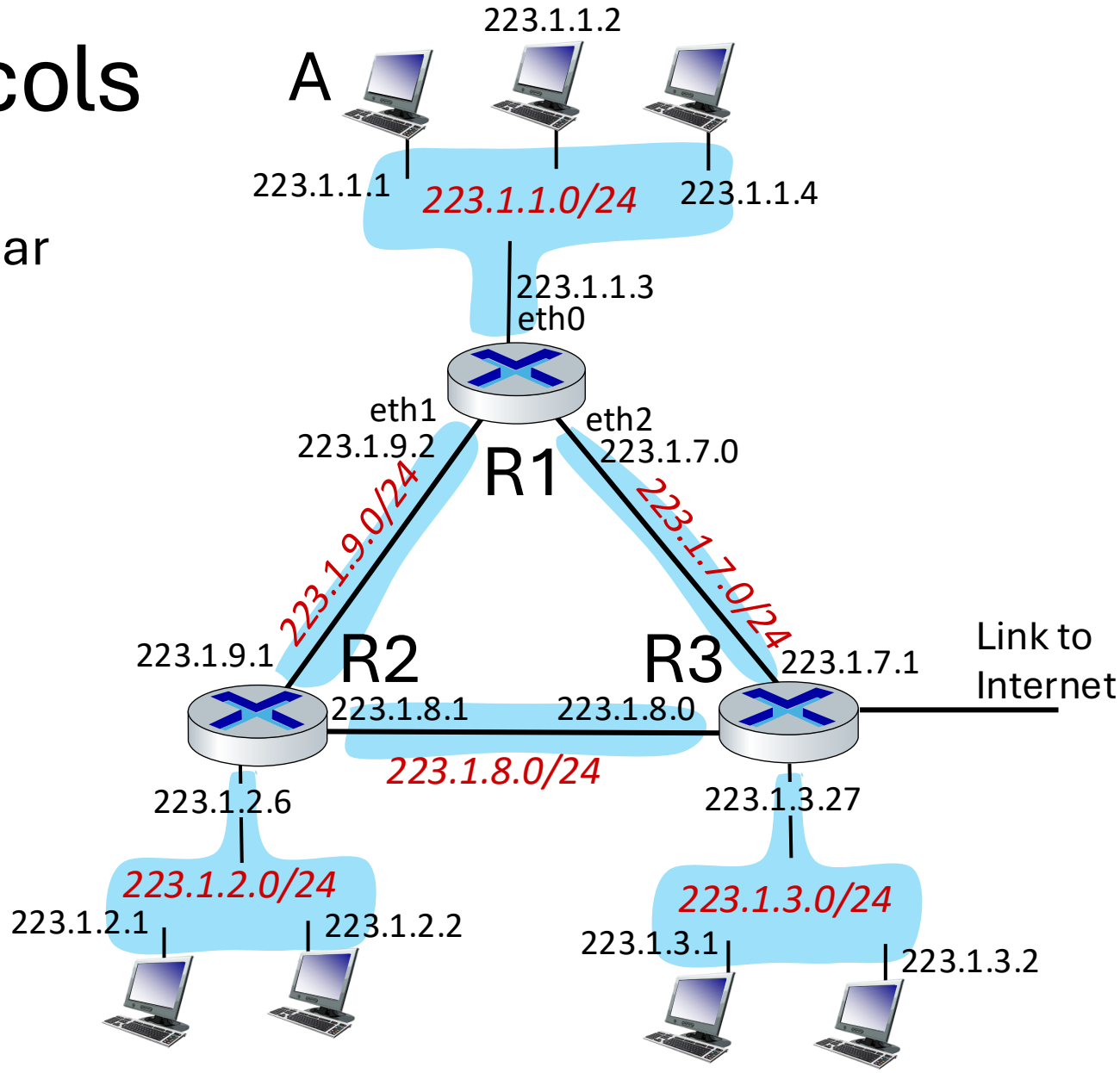


Distance vector protocols

- Each router tells other routers the shortest distance it has for a particular destination
- Example: R1 to others:

Destination	Distance
223.1.1.0/24	10
223.1.7.0/24	10
223.1.9.0/24	10
223.1.2.0/24	110
223.1.3.0/24	110
223.1.8.0/24	110

- Other routers use this information to update their routing tables
- Simple
- Bad news travels slowly (e.g. in RIP, a link going down would create a count-to-infinity problem)

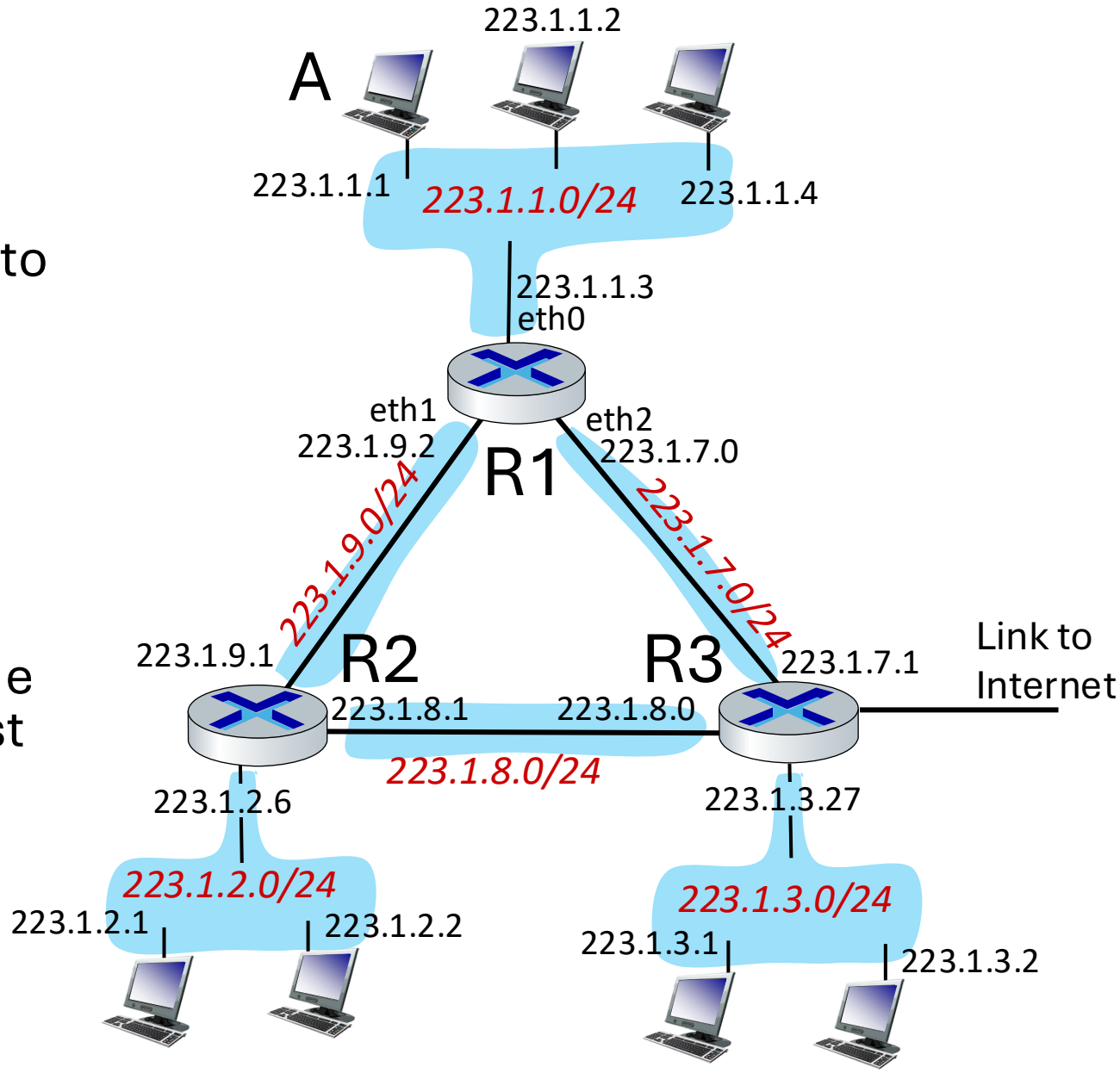


Link state protocols

- Each router tells others the state of each of the links that it is connected to
- Example: R1 to others:

Destination	Cost (metric)
223.1.1.0/24	5
223.1.7.0/24	5
223.1.9.0/24	5

- Other routers maintain a link state database which provides a map of the entire network, and compute the best path for destinations using some algorithm
- Requires more memory and processing power to compute paths
- Faster convergence



References

- Chapter 5 - The Network Layer: Control Plane
  James F. Kurose and Keith W. Ross, Computer Networking – A Top-Down Approach, (8th Edition), Pearson, 2020

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